

**PeriSense: AN AI-BASED MATERNAL RISK ASSESSMENT AND
DECISION-SUPPORT SYSTEM FOR IMPROVING MATERNAL HEALTHCARE
IN AFRICAN COUNTRIES**

BY

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DECLARATION

I, **Ademoye Ibraheem Aina**, a student of the Department of Computer Sciences, Faculty of Computing and Informatics, University of Lagos, with matriculation number **190805046**, under the supervision of **Dr. Babatunde Sawyerr**, declare that this report titled: **“PeriSense: An AI-Based Maternal Risk Assessment and Decision-Support System for Improving Maternal Healthcare in African Countries”** is the result of my original research, except where otherwise acknowledged. To the best of my knowledge and awareness, it has not been submitted, either in whole or in part, for any other academic degree or certification granted by this or any other higher learning institution. The information sources employed in this study are also adequately cited.

This report fulfills part of the requirements for earning the Bachelor of Science (B.Sc.) Degree in Computer Science from the University of Lagos, Lagos, Nigeria.

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CERTIFICATION

This is to certify that this project, titled “**PeriSense: An AI-Based Maternal Risk Assessment and Decision-Support System for Improving Maternal Healthcare in African Countries**”, was conducted by **Ademoye Ibraheem Aina**, a student of the Department of Computer Sciences, Faculty of Computing and Informatics, University of Lagos, bearing matriculation number **190805046**, under the guidance of **Dr. Babatunde Sawyerr**, as part of the requirements for obtaining the Bachelor of Science (B.Sc.) degree in Computer Science from the University of Lagos, Akoka, Yaba, Lagos, Nigeria.

This project has been reviewed and approved as fulfilling the criteria for the conferment of the degree.

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ABSTRACT

This study outlines **PeriSense**, an artificial intelligence-based maternal risk assessment and decision support system that will enhance maternal healthcare in African countries. Maternal morbidity and mortality remain a major problem in the public health of various low-resource regions, primarily because risks are often not identified at the earliest stage, quality healthcare remains insufficient, and the level of awareness about pregnancy-related complications is insufficient.

The purpose of this project is to design a predictive system that will utilize machine learning to determine the risk of maternal complications according to clinical and demographic characteristics. The system will generate customized risk categories and provide effective suggestions that can be acted upon to assist informative pregnant women and healthcare practitioners to make prompt decisions.

The method will include collecting information using the corresponding maternal health datasets, preparing information, selecting features, and using supervised machine learning algorithms to classify information. The performance of the model will be evaluated against the standard assessment tools, such as accuracy, precision, recall and F1-score. Besides, the system will entail basic explainability methods that will promote transparency and usability of clinical practices.

The expected future deliverable is a working web-based platform that combines predictive analytics with a friendly user interface that would help detect high-risk pregnancies and support early intervention. The research is an addition to the developing literature around

AI in the healthcare sector since it shows how machine learning can be used to forecast maternal risks in resource-constrained environments.

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CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Maternal health remains a significant issue of public health especially in the developing regions such as Africa where access to quality healthcare facilities is often limited. Global health reports indicate high levels of maternal death and complications during pregnancy because of avoidable conditions like hemorrhage, infections, hypertensive disorders, and the inability to access medical care in time (World Health Organization [WHO], 2019). Such difficulties are frequently associated with late diagnosis, the lack of proper healthcare facilities, the deficiency of qualified medical staff, and the lack of awareness among pregnant women (Say et al., 2014).

There is a chance to overcome these issues and enhance healthcare delivery through recent developments in artificial intelligence (AI) and machine learning (ML) as it allows predicting risks at an early stage and making a rational decision (Topol, 2019). Such technologies as machine learning models can be used to analyse a large amount of clinical and demographic data and reveal patterns and relationships that cannot be seen at first glance using more traditional approaches. Consequently, AI-driven systems will be able to help detect possible complications at an earlier stage and make more informed clinical decisions (Esteva et al., 2017).

PeriSense is suggested in this case as an AI-based maternal risk-assessment and decision-support system to enhance maternal healthcare in African countries. The system uses predictive analytics to determine the probability of complications and gives actionable information to guide healthcare workers and pregnant women to make informed choices in a timely manner.

1.2 STATEMENT OF THE PROBLEM

Nevertheless, in African contexts, even in the conditions of constant work to improve the situation in terms of maternal care, there is a high number of pregnant women who still encounter avoidable complications related to identification of potential risks and getting the necessary care in time (World Health Organization [WHO], 2020). Current medical methods are mainly response systems, where complications are being treated when they have reached other more severe stages.

Moreover, the absence of easily available and intelligent technology capable of aiding early risk detection and offering individual advice to medical personnel and pregnant women can be observed. The result of this gap is unhealthy, in addition to congesting the already stretched health care systems.

In this regard, a proactive, data-based solution is required that will enhance its capacity to detect risks early and be able to intervene in a timely manner.

1.3 AIM AND OBJECTIVES OF THE STUDY

1.3.1 Aim

This research will utilize the focus of developing an AI-driven maternal risk assessment and decision-support platform that will identify potential pregnancy complications and help improve healthcare outcomes in mothers.

1.3.2 Objectives

This research has specific research objectives, which are:

1. Create a machine learning-based solution in order to anticipate maternal health risks.
2. Identify and discuss the major clinical and demographic factors of maternal complications.
3. Create a web-based presentation of the risk assessment and recommendations.
4. Test the predictive model's performance with reasonable evaluation measures.
5. Add interpretability mechanisms to enhance the degree of the system transparency and usability.

1.4 SCOPE OF THE STUDY

This research will design a maternal risk prediction and decision-support platform through machine learning algorithms. The system will use available maternal health datasets that have proper clinical and demographic characteristics.

The scope includes pre-processing of data, model development, system design, and the implementation of a web-based user interface. Health practitioners and educated African women will be using the system.

The research, however, is not associated with clinical execution and integration with the hospital infrastructure in the present moment.

1.5 LIMITATIONS OF THE STUDY

The study has certain limitations to some extent, such as:

- Dependence on the access and the quality of secondary information
- The lack of local or real-time data
- Poor clinical validation due to academic expanse of the project
- Less complex recommendation procedure with real-world medical systems

1.6 METHODOLOGY

The methodology of this research is composed of some key steps such as the stage of data collection, preprocessing, model creation, system implementation, and evaluation.

Appropriate maternal health data will be acquired and it will be preprocessed to deal with missing values, feature normalization, and desired data to be trained. The performance metrics that are subject to be evaluated include accuracy, precision, recall, and F1-score, since predictive models are machine learning algorithms that will be determined through supervision.

The final model will be integrated in a web-based system through which the risk classification and simple decision support recommendation would be obtained.

1.7 SIGNIFICANCE OF THE STUDY

This study will add to the ongoing use of artificial intelligence in healthcare by illustrating how machine learning methods could be applied to avert the risk of the mother in resource-limited settings.

The solution suggested can:

- Increase the early intervention of maternal complications
- Aid the clinical decision-making for healthcare providers (Rajkomar et al., 2019)
- Enhance enlightenment among pregnant individuals

- Serve as a basis of subsequent research and possible application of the study in reality.

1.8 OUTLINE OF THE STUDY

Chapter 2 represents the literature review and discusses the existing works and relevant concepts to the area of study, namely; maternal healthcare, artificial intelligence in healthcare, machine learning to predict risks, and decision-support systems. It further determines gaps in existing studies that are filled in this study.

Chapter 3 entails the research methodology, comprising data sources, data preprocessing methods, model development strategy, and system architecture used in the entire research.

Chapter 4 outlines the implementation of the proposed system, tools, technologies, and processes that were involved in the development and integration of the predictive model and user interface.

Chapter 5 is where the system evaluation, results analysis and findings interpretation are given. This chapter also highlights the limitations, conclusion, and recommendations to the future research of the study.

CHAPTER TWO

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter provides a review of the literature available on maternal healthcare, artificial intelligence application in healthcare and the use of machine learning methods in predicting risks. It also reviews similar systems that have been developed due to the maternal risk assessment and identifies the gaps in the current research that the study will tackle.

2.2 MATERNAL HEALTHCARE IN AFRICAN COUNTRIES

Maternal health is a key issue in the overall health of the population especially in developing countries like Africa. The World Health Organization has reported that a disproportionate number of deaths associated with childbirth in the world are in the low- and middle-income nations, half of which is in sub-Saharan Africa (World Health Organization [WHO], 2019). Most of these deaths are as a result of complications which are avoidable like hemorrhage, hypertensive conditions, infections, and obstructed birth.

The factors that are likely to lead to poor maternal health outcomes are limited access to quality healthcare services, poor infrastructural conditions and lack of skilled healthcare professionals (Say et al., 2014). Besides, the socio-economic issue, poverty and lack of education contribute further to the problem as they delay the access to the medical care in due time.

Early recognition of high-risks pregnancy is crucial in lowering the morbidity and mortality in mothers. Nevertheless, it has been noted that in the African context, most of the healthcare systems depend on manual and reactive systems that are in most cases inadequate to respond in time.

2.3 ARTIFICIAL INTELLIGENCE IN HEALTHCARE

The advancement of artificial intelligence (AI) has emerged as a more significant instrument of contemporary healthcare because of its capacity to process and analyze extensive information. A subdivision of AI called machine learning (ML) allows systems to be trained based on historical data and make predictions or decisions without being explicitly programmed. With the help of machine learning algorithms, AI systems can identify the patterns that cannot be recognized by human professionals with ease and create information that can be used to assist in the diagnosis, prediction, and planning of treatment (Topol, 2019). These capabilities have at least been exploited in different realms of medicine, such as the diagnosis of diseases, patient monitoring, and predicting their risk, making them more accurate and making the care provided to the patients more personalized (Esteva et al., 2017).

Introducing AI into the healthcare systems has already proven useful in terms of efficient workflow, low human error, and making care more personalized. Nevertheless, the data quality, model interpretability, and ethics issues need to be overcome to provide safe and successful deployment.

2.4 MACHINE LEARNING FOR RISK PREDICTION

Machine learning applications are popular with predictive analysis in healthcare as they are able to model the complex relationship between variables as well as reveal the latent relationship among the variables within a given dataset. The commonly used classification algorithms include Logistic Regression, Decision tree, Random Forests, and Support Vector Machines that can be used to forecast the occurrence of health-related outcomes (Kourou et al., 2015).

Such models are conditioned on historical data containing the characteristics of patient demographics, medical history, and clinical measurements. The trained models can then be able to estimate the probability of events in the future and hence early intervention and better patient outcome.

Machine learning models in maternal healthcare have been investigated in the prediction of complications that include preeclampsia, gestational diabetes and preterm birth. These predictive systems can be used to assist the healthcare facilities to detect high-risk pregnancies and determine the right preventive actions.

Although these models are effective, not all machine learning models are interpretable because numerous complicated models may act as a “black box”, and clinicians may have a hard time learning how these predictions are formed.

2.5 DECISION-SUPPORT SYSTEMS IN HEALTHCARE

Clinical decision-support systems (CDSS) refers to computers that are utilized to assist healthcare providers in making informed decisions through support services, which are evidence-based suggestions and ideas (Shortliffe & Sepúlveda, 2018). These systems combine patient information with medical knowledge to aid the diagnosis, treatment organization and the risk evaluation.

The decision-support systems have been proven to enhance the healthcare quality by preventing the occurrence of diagnostic mistakes, increasing clinical effectiveness, and fostering the use of optimal practices. These systems can also be used in conjunction with machine learning models to offer personalized recommendations to individual data concerning a patient.

But effectiveness of decision-support systems is determined by certain factors like usability, accuracy, and trust. Healthcare professionals can be reluctant to use systems that are not very transparent or difficult to use.

2.6 RELATED WORKS

Some research has been conducted regarding the use of predictive models within the health care sector with specific focus directed to the risk identification of patients who face the danger of acquiring complications. As an example, one of the studies has investigated the use of statistical and machine learning models to predict the risk of maternal mortality and

has shown that the methods can enhance timely diagnosing and assist the clinical decision-making process (Caruana et al., 2015). There are however, several of these systems that are restricted by lack of accessibility, inadequate incorporation of decision-support functionalities, and the applicability of the systems to African healthcare settings.

Other researchers have paid attention to particular complications, including preeclampsia and preterm birth, and they employed predictive models to enhance early diagnosis and treatment (Oladapo et al., 2018). Although the research findings are encouraging, most of them are restricted to a particular area or data and may not be applicable to the African healthcare settings.

Moreover, not all the current systems have user-friendly interfaces as some are not as accessible to health care professionals and patients. Decision-support features are absent in others and do not give them much practical applicability in real-world contexts.

2.7 IDENTIFIED RESEARCH GAP

Based on the literature review, it is clear that although there has been a substantial improvement in the application of machine learning to the healthcare field, a good number of gaps remain:

- Little emphasis on maternal healthcare in African environs
- Absence of built-in systems which combine risk forecasting and disease support

- Lack of adequate focus on the interpretability of the model for clinical use
- Low accessibility and incompetence of the systems available

These loopholes confirm the need to have a complete system that will predict the dangers of motherhood as well as any other actionable information in a simple and easily accessible manner.

2.8 SUMMARY OF LITERATURE REVIEW

The chapter has discussed the available literature on maternal healthcare issues, the use of artificial intelligence in healthcare, machine learning methods of predicting risks, and decision-support systems. It has also reviewed similar works and found out the gaps that exist in present research.

The results of this review justify the development of **PeriSense**, an artificial intelligence-based maternal risk assessment and decision-support system that will help to eliminate these gaps and deliver more favorable outcomes of maternal healthcare in African environments.

CHAPTER THREE

METHODOLOGY

3.1 INTRODUCTION

This chapter outlines the methodology that was followed in the development of **PeriSense** such as research design, data sources, data preprocessing methods, machine learning model development, system architecture and the implementation tools. It will be aimed at offering a systematic way of developing an AI-based maternal risk assessment system and decision-support system.

3.2 RESEARCH DESIGN

In this research, a **quantitative, experimental research design** is utilized to create and test the machine learning models to identify maternal risks.

The approach used involves:

- Collecting and preparing maternal health records
- Training the predictive models and prejudging them
- Integrating the most successful model into a web-based platform

The research is a **data-based one**, and the results and prognosis are based on previous clinical and demographic data.

3.3 DATA SOURCES AND DESCRIPTION

The output and performance of any machine learning model will be dependent largely on the quality of the data as well as its relevance with which the machine learning model is trained. The data in this research paper will be obtained from **secondary sources**; this will include health related data on maternal health which are available in public repositories. In the dataset, there are clinical and demographic characteristics that are significant to ascertain the health risk of the mothers and they comprise:

- Reproductive age of the pregnant person
- Blood pressure levels
- Blood glucose levels
- Body mass index (BMI)
- Past pregnancies, and how many there are
- Medical history (if available)

The target variable would be the **risk level of the maternal complications**, and this may be measured as *low risk*, *medium risk*, and *high risk*.

3.4 DATA PREPROCESSING

One important step in the model training would be to preprocess the data to ensure that they are of good quality and reliable before the training is done.

The preprocessing methods that are used are:

- **Data Cleaning:** Processing of missing, mismatched or duplicate data values.
- **Data Transformation:** Scaling and organizing of the data to reduce redundancies and eliminate anomalies.
- **Encoding:** This is used to convert categorical variables into numbers.
- **Feature Selection:** This is necessary to determine those variables that are significant and impact the risk of the mother.
- **Data Splitting:** Dividing the data into training and testing sets (e.g., 80% training, 20% testing)

These measures help enhance the performance of the model and decrease the bias.

3.5 MODEL DEVELOPMENT

This study uses **supervised machine learning** to create the models that could predict the levels of maternal risk based on the input features. The algorithms chosen go through training of labelled data where the input features match the known results. It is aimed at

making the models learn underlying patterns and properly classify new cases on the basis of similarities.

The models put into consideration include the following:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

Each model is trained on the prepared dataset and assessed based on its predictive power in inferring maternal risk (Kourou et al., 2015).

3.6 MODEL EVALUATION

To assess the performance of the models, the evaluation measures used are as follows:

- **Accuracy:** Determines how correct the predications are.
- **Precision:** Measures the relative number of correctly predicted positive cases.
- **Recall (Sensitivity):** Tests the real cases from the positive cases.
- **F1-Score:** Used to balance between precision and recall.

A confusion matrix is also used to represent the performance of each model.

The model with the best performance based on these measures is chosen and incorporated into the system.

3.7 SYSTEM DESIGN

The PeriSense system is a **web-based application**, which incorporates both a machine learning model with a user-friendly interface.

3.7.1 System Architecture

The system has a **three-tier architecture**, consisting of:

1. **Presentation Layer (Frontend):**

Provides the interface through which interaction with the user is to take place, such as dashboards and input forms.

2. **Application Layer (Backend):**

Processes business logic, API requests, authentication, and communication with the machine learning model.

3. **Data Layer (Database):**

Stores user data, medical records, and prediction results.

3.7.2 System Workflow

The workflow of the system is broken down into the following:

1. User inputs the relevant data for maternal healthcare
2. Data is sent to the backend server via API to carry out the analysis
3. The machine learning model processes the input and uses it

4. The system creates a risk classification
5. Results and recommendations are then presented to the user.

3.8 DECISION-SUPPORT MECHANISM

The system contains a rudimentary **decision-support component**, in addition to prediction.

The system has a decision-support feature, alongside risk prediction (low, medium, and high) that assists the user to make interpretation on the results. Depending on the level of risk as predicted by the system, the system will offer basic guidelines to direct the users on the correct course of action which may include regular monitoring, a check up with a doctor, or hospitalization. The suggestions are guideline-based and comply with universal guidelines for healthcare provided to mothers (Shortliffe & Sepulveda, 2018).

3.9 SYSTEM IMPLEMENTATION TOOLS

The development of the PeriSense system uses the following technologies:

3.9.1 Frontend

- React.js to develop the user interface
- Redux for state management
- Axios for API communication

3.9.2 Backend

- C# with ASP.NET for server-side logic
- JWT for authentication and authorization

3.9.3 Database

- Azure SQL as a data storage facility

3.9.4 Machine Learning

- Azure AI / ML tools for model training and deployment

3.9.5 Testing

- Xunit for unit testing
- FakeItEasy for mocking
- Postman for API testing

3.10 ETHICAL CONSIDERATIONS

Given that no healthcare data can be taken as not sensitive, the following are the ethical aspects that have been put into consideration in the course of the study:

- Protecting information privacy and confidentiality
- Avoiding the use of personally identifiable information
- Using datasets strictly for research purposes
- Giving non-diagnostic advice (not to the exclusion of the medical professionals)

3.11 SUMMARY

The chapter has described the procedure followed in the development of the PeriSense system such as the collection of data, its preprocessing, model development, evaluation and system design.

The next chapter will dwell upon the **system implementation**, which will include how the proposed system will be implemented on the basis of the chosen tools and technologies.

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